

BiLSTM Speech Enhancement Pipeline

ATSP Project Report

Deep Learning Audio Processing and Speech Enhancement

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Abstract

This project focuses on building and improving a deep learning-based speech enhancement system using a Bidirectional Long Short-Term Memory (BiLSTM) model. The goal of the system is to separate speech from noise by predicting a mask in the time-frequency domain rather than generating speech directly.

Throughout development, we explored both the mathematical concepts behind speech enhancement and the practical engineering challenges involved in making the system work on real-world audio. Initial experiments worked on synthetic datasets, but several issues such as reverberation artifacts, robotic sound quality, dataset mismatch, and memory bottlenecks appeared during testing.

This report explains the design of the pipeline, the reasoning behind major technical choices, the problems we faced, and the solutions we implemented. We also discuss future improvements, especially replacing memory-heavy data loading with generator-based training for larger datasets.

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1. Introduction

Speech enhancement is the process of improving the quality of speech by reducing background noise. This is useful in many applications such as voice assistants, online meetings, hearing aids, and automatic speech recognition systems.

In this project, we built a BiLSTM-based speech enhancement model that works in the frequency domain. Instead of processing raw waveforms directly, we first convert audio into spectrogram-like representations using the Short-Time Fourier Transform (STFT). The model then learns to predict which parts of the signal belong to speech and which belong to noise.

Our project involved:

- Understanding STFT and spectral masking,
- Building a BiLSTM model,
- Training on noisy-clean speech pairs,
- Debugging reconstruction artifacts,
- Improving scalability for larger datasets.

2. Time-Frequency Representation

2.1 Why Not Use Raw Audio?

Raw audio contains thousands of samples per second. For example, at 16 kHz, one second of audio has 16,000 values. Feeding such long sequences directly into an LSTM is computationally expensive.

To simplify this, we convert the audio into smaller overlapping frames using the Short-Time Fourier Transform.

2.2 Short-Time Fourier Transform (STFT)

The STFT is defined as:

$$X[k, m] = \sum_{n=0}^{N-1} x[n + mH] \cdot w[n] \cdot e^{-j\frac{2\pi kn}{N}} \quad (1)$$

where:

- N = FFT size,
- H = hop size,

- $w[n]$ = window function,
- k = frequency bin,
- m = time frame.

For our project:

- FFT size = 512,
- Hop length = 128,
- Sampling rate = 16 kHz.

This transforms the audio into a 2D matrix where one axis is time and the other is frequency.

3. Ideal Ratio Mask (IRM)

Instead of predicting clean speech directly, our model predicts a mask:

$$M(k, m) = \frac{|S(k, m)|}{|Y(k, m)| + \epsilon} \quad (2)$$

where:

- $|S(k, m)|$ = clean speech magnitude,
- $|Y(k, m)|$ = noisy speech magnitude,
- ϵ = small constant to avoid division by zero.

The mask tells us how much of each frequency bin should be kept. Values close to 1 mean speech is dominant, while values close to 0 mean noise should be suppressed.

4. BiLSTM Model Architecture

4.1 Why BiLSTM?

Speech depends on context. Understanding a sound often requires information from both past and future frames.

A BiLSTM processes the sequence in two directions:

- Forward pass,
- Backward pass.

This allows the model to capture better temporal information.

Forward:

$$\vec{h}_t = \mathcal{H}(W_{\vec{h}}x_t + U_{\vec{h}}\vec{h}_{t-1}) \quad (3)$$

Backward:

$$\overleftarrow{h}_t = \mathcal{H}(W_{\overleftarrow{h}}x_t + U_{\overleftarrow{h}}\overleftarrow{h}_{t+1}) \quad (4)$$

Combined:

$$H_t = [\vec{h}_t; \overleftarrow{h}_t] \quad (5)$$

4.2 Output Layer

The hidden representation is passed through a dense layer:

$$Z_t = H_t W + b \quad (6)$$

Then sigmoid activation:

$$\hat{M}_t = \frac{1}{1 + e^{-Z_t}} \quad (7)$$

This ensures mask values remain between 0 and 1.

5. Training Process

We used Mean Squared Error (MSE) loss:

$$\mathcal{L} = \frac{1}{N} \sum (M - \hat{M})^2 \quad (8)$$

The optimizer used was Adam:

$$W_{t+1} = W_t - \frac{\alpha m_t}{\sqrt{v_t} + \epsilon} \quad (9)$$

Adam helped training converge faster compared to standard gradient descent.

6. Problems Faced and Solutions

6.1 Temporal Smearing

Initially, the output sounded reverberant. This happened because our frame overlap was too high, causing the same STFT frame to be predicted too many times.

Fix: Increased FRAME_STEP from 1 to 16.

6.2 Robotic or Hollow Audio

Since we only predicted magnitude and reused noisy phase, the enhanced speech sounded unnatural.

Fix: Added a spectral floor:

$$\text{Mask} = \max(M, 0.05) \quad (10)$$

6.3 Synthetic Dataset Issue

Training on sine waves did not generalize to real human speech.

Fix: Shifted focus to real-world speech datasets.

7. Scalability Improvements

Originally, all files were loaded into RAM using `numpy.concatenate`, which worked for small datasets but failed for larger ones.

7.1 Future Plan

We plan to implement a `tf.keras.utils.Sequence` generator:

1. Load one file,
2. Process it,
3. Train,
4. Clear memory,
5. Repeat.

This will allow us to train on thousands of files without memory crashes.

8. Conclusion

This project helped us understand both the theoretical and practical aspects of speech enhancement. Beyond building the BiLSTM model, we learned that engineering decisions such as framing strategy, dataset quality, reconstruction methods, and memory management are equally important.

Our current system works as a strong foundation, and future improvements such as real-world datasets and generator-based training can make it more robust and scalable.

References

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